
SWEDISH PATENT SE 301 717 B

Method for Producing Thermoplastics from a Dihydroxy Compound and a Diepoxy Compound, Wherein the Diepoxy Compound or Both Compounds Contain an SO₂ Group

English translation and clean typeset edition

INVENTORS

R. W. Kreps • Arie Klootwijk • J. M. Goppel

APPLICANT

Shell Internationale Research Maatschappij N.V., The Hague

PRIORITY / PUBLICATION

15 June 1959 • 17 June 1968

Prepared from the original Swedish publication supplied by the Klootwijk family.
This is an unofficial translation and is not a certified legal text.

Publication Details

Authority	Swedish Patent and Registration Office
Publication	Laid-open specification No. 301 717
Original Swedish title	Sätt att framställa termoplaster av en dihydroxiförening och en diepoxiförening varvid diepoxiföreningen eller båda innehåller en SO ₂ -grupp
Classification	International Class C 08 g 29/04; Swedish Class 39 c 30
Application	No. 5843/60; filed 15 June 1960
Effective date	15 June 1960
Public availability	Application made publicly available 17 June 1968
Publication date	Application laid open and specification published 17 June 1968
Priority	Netherlands No. 240 201, dated 15 June 1959
Applicant	Shell Internationale Research Maatschappij N.V., The Hague, Netherlands
Inventors	R. W. Kreps; Arie Klootwijk; J. M. Goppel
Representative	E. Lindquist

About This Edition

This edition translates the complete Swedish description, six working examples, reconstructed results tables, all eleven patent claims, and the cited publication. The original order and technical level have been preserved, while spelling, punctuation, paragraphing, chemical notation, and table layout have been regularized for readability.

EDITORIAL AND TRANSLATION NOTES

The title above follows the Swedish title as printed. In the body and in Claim 1, the text states more broadly that either the dihydroxy compound, the diepoxy compound, or both may contain the SO₂ group.

On original page 3, the first endpoint of the molar-ratio range appears in the scan as “1.4”. The corresponding patent-family text gives “1.04”, which is consistent with the preferred range immediately following it. This edition therefore uses 1.04 and records the correction here rather than silently altering the source.

Decimal commas have been converted to decimal points. Obvious typographical inconsistencies in repeated chemical names have been normalized, including omissions of “ol” in “diphenylol sulfone”. Historical compound names and the published units have otherwise been retained.

To resolve scan ambiguities, the Swedish text was cross-checked against corresponding family publications GB 915,767 A and DE 1,190,672 B. The Swedish publication remains the basis of this translation.

Description

Field of the Invention

The invention relates to a method of producing thermoplastics suitable for the manufacture of threads, fibres and the like. These thermoplastics have a high molecular weight and an almost exclusively linear structure. They are obtained by reacting one or more optionally substituted dihydroxy compounds with one or more optionally substituted diepoxy compounds, wherein either the dihydroxy compound, the diepoxy compound, or both contain an SO₂ group joining two aromatic nuclei, preferably a -C₆H₄-SO₂-C₆H₄- group.

Background

A method falling within the scope described above is already known from British Patent No. 652,024, particularly from page 7, line 125, to page 8, line 51. A two-stage process is disclosed there. In the second stage, a dihydroxy compound whose hydroxyl groups are attached to aromatic nuclei is reacted with an equimolar amount of a diglycidyl ether of such a dihydroxy compound in the presence of a small amount of an alkaline substance as catalyst. Diphenylol sulfone is mentioned elsewhere in that patent specification as a dihydroxy compound.

The relevant portion of that patent specification also permits a solvent to be added together with the diglycidyl ether. Only ethanol and dioxane are mentioned for this purpose. When diphenylol sulfone and/or the diglycidyl ether derived from it are used as starting materials, however, less satisfactory products are obtained in this way. It has now been found that this less satisfactory result is caused by premature interruption of the reaction through crystallization of the product, or by the solvent present acting as a chain-terminating agent.

It has been found that products having higher molecular weights and substantially improved properties are obtained when this premature interruption of the reaction is prevented.

The Invention

According to the invention, thermoplastics are produced by reacting one or more optionally substituted dihydroxy compounds with a substantially equimolar amount of one or more optionally substituted diepoxy compounds. Either the dihydroxy compound, the diepoxy compound, or both contain an SO₂ group joining two aromatic nuclei. The reaction is carried out in the presence of a solvent and an alkaline compound as catalyst. The process is characterized in that the reaction takes place in a solvent in which the dihydroxy compound, the diepoxy compound and the reaction product dissolve, and which scarcely reacts with the dihydroxy compound or the diepoxy compound.

Suitable Solvents

Solvents suitable for use according to the invention generally have a dielectric constant of at least 5 at the reaction temperature. Solvents having a dielectric constant of at least 10 at the reaction temperature usually give the best results.

Examples of suitable solvents are ketones such as methyl isobutyl ketone and cyclohexanone, particularly aromatic ketones such as acetophenone and benzophenone; aldehydes such as furfural; nitriles such as acetonitrile, phenylacetonitrile, propanedinitrile and benzonitrile; nitro compounds such as nitrobenzene, 1-chloro-2-nitrobenzene and 1-chloro-3-nitrobenzene; organic sulfoxides such as dimethyl sulfoxide; and sulfones such as cyclotetramethylene sulfone (sulfolane), 3-methylcyclotetramethylene sulfone and 3,4-dimethylcyclotetramethylene sulfone.

Especially important sulfoxo compounds and nitriles are principally those in which a divalent sulfoxo group (SO or SO₂), or a cyano group, is directly attached to at least one carbon atom which is itself directly attached to one or more hydrogen atoms. If desired, the solvent may be a mixture of different components, for example a mixture of di-n-propyl sulfoxide and diphenyl sulfoxide. Solvents that catalyse side reactions, for example reactions of the epoxy groups, are undesirable.

Reactant Ratio and Mode of Addition

The expression "substantially equimolar amounts" means quantities in which the molar ratio of dihydroxy compound to diepoxy compound is from about 1.04 to about 0.98, preferably from about 1.02 to about 0.99. Any amount of dihydroxy compound present as a salt, for example an alkali-metal salt of a dihydric phenol, is included in this ratio as dihydroxy compound.

The diepoxy compound may be added gradually to the dihydroxy compound, as indicated in the British patent. It has been found, however, that this precaution is unnecessary in the process according to the present invention. There is no objection to mixing, in a single operation, the entire quantities of the diepoxy and dihydroxy compounds intended for reaction, provided that those quantities are measured with the required degree of accuracy.

Alternatively, the diepoxy compound may first be reacted with a slight excess of the dihydroxy compound, for example an excess of a few percent, and the ratio between the compounds may be corrected to an equimolar ratio at a later stage of the reaction.

Preferred Compounds

The preferred dihydroxy compound is a dihydric phenol in which the hydroxyl groups, when attached to the same aromatic nucleus, are not in ortho positions relative to one another.

Preferred diepoxy compounds are the di(epoxyalkyl) ethers of the dihydric phenols mentioned above, and especially the diglycidyl ethers.

Very good products are obtained by reacting diphenylol sulfone with its diglycidyl ether. Products obtained according to the invention by reacting diphenylol sulfone with the diglycidyl ethers of other dihydric phenols, or by reacting those other dihydric phenols as such with the diglycidyl ether of diphenylol sulfone, are likewise very suitable.

Examples of these other dihydric phenols are, in the first instance, 2,2-diphenylolpropane, and also diphenylolmethane, hydroquinone, resorcinol, 3,3'- and 4,4'-dihydroxydiphenyl, various dihydroxydiphenoxyethanes, and various dihydroxynaphthalenes. Other suitable starting materials are mixtures containing diphenylol sulfone and its diglycidyl ether together with one or more of these other dihydric phenols and/or their glycidyl ethers.

Molecular Weight and Product Properties

The beneficial effect of the solvent used according to the invention is due to the product remaining in solution until its molecular weight has risen to relatively high values. A suitable choice of solvent, solvent quantity and temperature makes it possible to avoid separation of a solid macromolecular product, in either crystalline or amorphous form, until the product has reached an intrinsic viscosity of at least about 0.4, preferably at least about 0.8, at reasonable product concentrations in the reaction mixture, for example at least 5 wt% and preferably at least 10 wt%. In this way, much higher intrinsic viscosities, for example 1.4, can even be obtained. This is shown more clearly in the examples below. All intrinsic-viscosity values stated were determined in dimethylformamide.

The increase in molecular weight is accompanied by improved mechanical properties. Very good values are obtained for yield stress and impact strength. The products also have the advantages of a high softening point and a high heat-distortion temperature, together with strong chemical resistance to acids and bases.

The British patent referred to above describes how products of the intended kind may be drawn or spun into threads. The disclosure of that patent applies, *mutatis mutandis*, when the products are made according to the present invention. The products of this invention are also suitable for making articles other than threads by operations such as casting, extrusion through a die, injection moulding, rolling and the like.

Water Content and Catalysts

Among the conditions under which the reaction is allowed to proceed, the advantages of excluding water and using pure starting materials should also be mentioned.

Preference for an anhydrous or substantially anhydrous medium means that the catalyst is preferably also introduced in an anhydrous state and that the added catalyst is a substance which does not cause water to be formed in the reaction mixture. Although alkali-metal hydroxides of the kind used in the British patent are also suitable for the present invention, they have the disadvantage that they form water when they react with phenols.

It is therefore preferred to add separately prepared salts of strong bases and dihydric phenols to the reaction mixture. Examples of such strong bases are bases of the alkali metals and alkaline-earth metals, and quaternary ammonium bases. These salts are naturally prepared, preferably, from the dihydric phenol taking part in the reaction. Monosalts of dihydric phenols give very good results.

A suitable method of preparing such salts in an anhydrous state is to react the dihydric phenol with an alkali-metal alkoxide and distil off the alcohol liberated.

The permissible limits for water content depend on the requirements to be met by the product and on the nature of the medium. It is advisable to keep the water content below 2 wt%, preferably below 1 wt%, and especially below 0.3 wt%, calculated on the original dihydric phenol.

Salts of strong bases and dihydric phenols are generally used in amounts from 0.01 to 10 mol%, especially from 0.1 to 5 mol%, calculated on the dihydric phenol originally present.

Salts of acids other than phenols, for example the stearates of magnesium, aluminium and trivalent iron, may likewise be added as catalysts when they are sufficiently soluble in the reaction mixture.

Purification of Starting Materials

Commercial diphenylol sulfone generally contains so many impurities that it must be purified before use. The impurities are partly soluble in water and partly soluble in aromatic solvents such as benzene and toluene, in which diphenylol sulfone itself has only slight solubility. Suitable purification methods are therefore extraction with water and aromatic solvents, preferably at elevated temperature.

Diphenylol sulfone may be crystallized by dissolving it in a ketone, such as acetone or methyl isobutyl ketone, and adding an aromatic compound such as benzene to the solution. This method is particularly suitable when water-soluble impurities are absent, or after they have otherwise been removed.

A new purification method consists in recrystallizing diphenylol sulfone from water at temperatures above 100°C. The solubility curve, which is very low and flat below 100°C, shows a marked rise above that temperature, especially between 100 and 125°C. To remove impurities soluble in aromatic compounds at the same time, this recrystallization is carried out from water to which a small amount of an aromatic compound has been added.

Recrystallization also reduces the amount of o,p'-diphenylol sulfone present alongside the p,p'-diphenylol sulfone and consequently gives a macromolecular end product of better quality. For the reaction between diphenylol sulfone and its diglycidyl ether, a diphenylol sulfone having a melting point of at least 240°C is preferred as starting material.

The diglycidyl ether of diphenylol sulfone generally has a sufficient degree of purity when its melting point reaches 162°C. It is also advisable to ensure that this ether contains little or no chloride, for example not more than 0.5 wt%. If necessary, the purity of the diglycidyl ether of diphenylol sulfone may be increased by recrystallization, for example from a ketone such as methyl isobutyl ketone.

Reaction Conditions

The temperature at which a dihydroxy compound is reacted with a diepoxy compound according to the invention generally lies between 80 and 200°C, although it may be lower or higher in particular cases. If necessary, pressure is used to keep the solvent in the liquid state during the reaction. It is advisable to conduct the reaction in the absence of oxygen.

ORIGINAL SWEDISH PAGES 5-8

Working Examples

The following examples illustrate the invention. Intrinsic viscosity was measured in dimethylformamide unless otherwise stated.

Example 1

A mixture of 1.00 mol of the diglycidyl ether of p,p'-diphenylol sulfone (melting point 166.6-176.4°C) and 0.97 mol of p,p'-diphenylol sulfone (melting point 248.5-249.5°C) was dissolved in 850 g of acetophenone in a round-bottom flask fitted with a thermometer, stirrer, reflux condenser and gas-inlet tube. Thereafter, 0.05 mol of the monosodium salt of p,p'-diphenylol sulfone was added.

Oxygen was removed from the flask atmosphere by passing in nitrogen. The mixture was then heated to 100°C and maintained at that temperature for 60 hours with stirring. It was then cooled to room temperature. Crystalline material had already begun to separate during the reaction, and crystallization of the product was complete a few hours after the reaction ended. The crystalline material was removed by suction filtration, washed with methanol and dried at 100°C in a vacuum desiccator.

The product had an intrinsic viscosity of 0.43 and a softening point of 155°C. Birefringence and a sharp X-ray diffraction pattern were also observed. The birefringence disappeared on heating to about 230°C, the melting point of the crystallites.

Small plates were pressed from the crystalline powder at temperatures in the range 230-250°C. Pressing caused no difficulty because the product showed excellent flow after softening. Properties were measured at room temperature. Yield stress was also measured after heating for selected periods at 160°C in order to assess thermal stability. Performance under load was evaluated by determining the heat-distortion temperature according to the ASTM standard. The effects of water and 5% aqueous solutions of sodium hydroxide and sulfuric acid were studied by prolonged immersion of the plates. The liquids were absorbed, but no chemical change occurred. The very satisfactory results are summarized below.

Property	Result
Yield stress	770 kg/cm ²
Elongation at yield stress	15-20%
Impact tensile strength	650 kg·cm/cm ²
Izod impact strength (British Standard)	30 kg·cm/cm ²
Heat-distortion temperature, ASTM D 648, at 18.9 kg/cm ² load	154°C

Yield Stress after Heating at 160°C

Heating time	Yield stress
0 hours	770 kg/cm ²
24 hours	1,000 kg/cm ²
72 hours	1,000 kg/cm ²
100 hours	1,000 kg/cm ²

Weight Gain at 20°C

Immersion medium	1 day	7 days	14 days
Distilled water	3.4 wt%	5.3 wt%	5.6 wt%
5 wt% NaOH solution	3.2 wt%	4.8 wt%	5.1 wt%
5 wt% H ₂ SO ₄ solution	3.3 wt%	5.1 wt%	5.4 wt%

Example 2

The reaction was carried out in the same manner as in Example 1, except that the mixture was maintained at 100°C for only 20 hours. The intrinsic viscosity of the crystalline product was 0.35.

Small plates were pressed from the crystalline powder at 230°C. This treatment did not affect the intrinsic viscosity. The pressed material had a yield stress of 750 kg/cm², an elongation at yield stress of 15-20%, and an impact tensile strength of 110 kg·cm/cm².

Example 3

The reaction was carried out in the same manner as in Example 1, except that during the experiment the temperature was raised to 150°C in order to redissolve the crystallizing product. Two samples taken during the experiment showed intrinsic viscosities of 0.45 and 0.50, respectively. Further polymerization in these samples was prevented by adding small amounts of p,p'-diphenylol sulfone.

After plates had been pressed from the two samples, at 230°C and 250°C respectively, the following results were obtained:

Property	Sample 1	Sample 2
Intrinsic viscosity	0.46	0.50
Yield stress	770 kg/cm ²	775 kg/cm ²
Elongation at yield stress	15-20%	15-20%
Impact tensile strength	650 kg·cm/cm ²	620 kg·cm/cm ²

Comparison of these values with the results of Example 2 showed that the mechanical properties depended on the molecular weight of the product, as represented by its intrinsic viscosity. Good impact strength was obtained only at intrinsic viscosities exceeding 0.35.

Example 4

The reaction was carried out as described in Example 1, except that the reaction temperature was 150°C and the reaction time was 50 minutes. After that period, a solution of 2.5 g of p,p'-diphenylol sulfone in 25 g of acetophenone was added to stop the reaction. The mixture was maintained at 150°C for a further 15 minutes, then cooled to 100°C and diluted with dimethylformamide to give a 10% solution of the polymer.

Methanol was added to this solution at room temperature with stirring until the polymer had completely precipitated. The precipitate was washed with methanol and redissolved in dimethylformamide, this time at a concentration of 2%. Methanol was added to the new solution until slight cloudiness appeared, after which the solution was poured into distilled water with vigorous stirring. The polymer flocculated as small particles. These were removed by suction filtration and washed with water. The resulting powder was dried at 100°C in a vacuum desiccator. The product had an intrinsic viscosity of 1.37.

Small plates were pressed from the crystalline powder at 290°C. This treatment reduced the intrinsic viscosity to 1.29. The material pressed into plates had a yield stress of 775 kg/cm², an elongation at yield stress of 15-20%, and an impact tensile strength of 690 kg·cm/cm².

Example 5

A mixture of 1.00 mol of the diglycidyl ether of p,p'-diphenylol sulfone and 1.00 mol of p,p'-diphenylol sulfone was combined with 125 g of sulfolane (tetramethylene sulfone), in which 0.002 mol of the monosodium salt of p,p'-diphenylol sulfone had been dissolved. The mixture was melted and maintained in a sealed tube, with air excluded, at 105°C for 24 hours. The resulting polymer had an intrinsic viscosity of 0.42.

Example 6

A quantity of 1.00 mol of the diglycidyl ether of p,p'-diphenylolpropane (melting point 42°C) was reacted with 0.97 mol of p,p'-diphenylol sulfone in the presence of 0.05 mol of the monosodium salt of p,p'-diphenylol sulfone, in exactly the same manner as in Example 4. The product was also isolated in the manner described in Example 4. Its intrinsic viscosity was 0.82.

Property	Result
Yield stress	770 kg/cm ²
Izod impact strength (British Standard)	33 kg·cm/cm ²
Impact tensile strength	300 kg·cm/cm ²
Heat-distortion temperature, ASTM D 648, at 18.9 kg/cm ² load	114°C
Water absorption after 140 hours at 20°C	1.8 wt%
Glass-transition point from modulus-temperature relationship	about 130°C

ORIGINAL SWEDISH PAGES 9-10

Patent Claims

The following claims are translated in their original sequence. Patent-style dependency language has been retained.

1. A method of producing thermoplastics by reacting one or more dihydroxy compounds with a substantially equimolar amount of one or more diepoxy compounds, wherein either the dihydroxy compound, the diepoxy compound, or both contain an SO₂ group joining two aromatic nuclei, in the presence of a solvent and an alkaline compound as catalyst, characterized in that the reaction is carried out in a substantially anhydrous ketone, nitrile, nitro compound, sulfoxide or sulfone having a dielectric constant of at least 5 as solvent, in which the dihydroxy compound, the diepoxy compound and the reaction product are soluble and which does not react with the dihydroxy compound or the diepoxy compound, and in that the reaction is allowed to continue until the polymeric reaction product has reached an intrinsic viscosity of at least 0.4.
2. A method according to Claim 1, characterized in that the dihydroxy compound is a dihydric phenol in which the hydroxyl groups, when attached to the same aromatic nucleus, are not in ortho positions relative to one another.

3. A method according to Claim 1 or 2, characterized in that the diepoxy compound used is a di(epoxyalkyl) ether of a dihydric phenol in which the ether oxygen atoms are not in ortho positions relative to one another in the same aromatic nucleus.
4. A method according to any preceding claim, characterized in that the dihydric phenol is p,p'-diphenylol sulfone and the diepoxy compound is the diglycidyl ether of p,p'-diphenylol sulfone.
5. A method according to Claim 1, characterized in that the nature and quantity of the solvent and the temperature are such that the reaction product remains dissolved in the reaction mixture until an intrinsic viscosity of at least 0.8 has been reached.
6. A method according to any preceding claim, characterized in that the solvent used is a substance having a dielectric constant of at least 10.
7. A method according to any preceding claim, characterized in that the reaction mixture contains a substantially anhydrous catalyst.
8. A method according to any preceding claim, characterized in that the catalyst used is a salt of the dihydric phenol which takes part in the reaction.
9. A method according to Claim 8, characterized in that the catalyst used is a mono-alkali-metal salt of a dihydric phenol.
10. A method according to any preceding claim, characterized in that the starting material used is a diphenylol sulfone having a melting point of at least 240°C.
11. A method according to any preceding claim, characterized in that the starting material used is a diglycidyl ether of a diphenylol sulfone, which ether has a melting point of at least 162°C.

Cited Publication

United States Patent No. 2,767,157 (class 260.47).

END OF TRANSLATION

This clean English edition corresponds to the complete ten-page Swedish publication SE 301 717 B. The original facsimile should be consulted for any legal or historical use requiring the exact Swedish wording.